

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	15 February 2026
Team ID	LTVIP2026TMIDS82308
Project Name	DocSpot: Seamless Appointment Booking for Health
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for patients, doctors, and admin dashboards	React.js, HTML, CSS, Bootstrap, Material UI
2.	Application Logic-1	Authentication & Role Management	Node.js, Express.js, JWT
3.	Application Logic-2	Appointment Booking & Scheduling	Express.js REST APIs
4.	Application Logic-3	Notification & Status Management	Node.js Controllers
5.	Database	Stores users, doctors, appointments data	MongoDB, Mongoose
6.	Cloud Database	Cloud-hosted NoSQL database for storing users, doctors, appointments, and notifications with remote access and scalability	MongoDB Atlas
7.	File Storage	Medical document upload & storage	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	HTTP communication between frontend & backend	Axios
9.	External API-2	Not Applicable	—

10.	Machine Learning Model	Not used in this project	—
11.	Infrastructure (Server / Cloud)	Local development deployment	Node.js Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build UI & Backend APIs	React.js, Express.js, Node.js
2.	Security Implementations	Role-based authentication, encrypted passwords, protected routes	JWT, bcryptjs, Middleware
3.	Scalable Architecture	3-Tier architecture separating UI, backend logic, and database	REST Architecture, MongoDB
4.	Availability	Web application accessible anytime through browser	Node.js Server
5.	Performance	Fast API responses and asynchronous communication	Axios, Express.js